



SHELL'S PATH TO SUSTAINABLE OFFSHORE REPURPOSING

2024 USYD X MONASH MCKINSEY CASE COMPETITION



CORE PROBLEM

FINANCIAL CHALLENGES

Cost and benefits of decommissioning vs repurposing.



ENVIRONMENTAL CONSIDERATIONS

Ecological considerations when comparing benefits vs negative effects of structures.



REGULATORY REQUIREMENTS

Case-by-case evaluation for decommissioning project and compliance with OSPAR Decision 98/3.



Multi-option Evaluation Screening Matrix

1 = low, 10 = high (of preference)
*considering risks and barriers involved
using sample data from the case with
regulatory being averagely assumed*



	Financial	Environmental	Regulatory	Total Score
Potential Repurposed Technologies	- Existing infrastructure - Levelised cost - Capital cost of energy tech	- GHG Emissions - Ecosystem impacts	- Subsidies - Regulatory complexity - Safety regulations	
Carbon Capture and Storage (CCS) Captures and stores carbon dioxide from emissions to reduce atmospheric pollution	3	2	1	6
Hydrogen Generation (Green) Produces hydrogen using renewable energy with zero carbon emissions	8	9	2	19
Hydrogen Generation (Blue) Produces hydrogen from natural gas with carbon capture to minimise emissions	9	3	2	14
Hydrogen Generation (Turquoise) Produces hydrogen by splitting methane into solid carbon and hydrogen	4	5	2	11
Hydrogen Generation (Grey) Produces hydrogen from natural gas without capturing carbon emissions	8	3	2	13
Offshore Wind Farms Utilises wind turbines placed offshore to generate renewable electricity	4	9	3	16

Cost & Benefit Analysis



DECOMMISSIONING

REPURPOSING

COST

Direct

- Estimated AUD 77-154 million per platform, Infrastructure removal, disposal, environmental compliance.

Indirect

- Economic impact on local communities, potential environmental liabilities.

Direct

- Retrofitting costs, Initial investment in technology and infrastructure, cost of producing green hydrogen (LCOH).

Indirect

- Operational expenses, Potential Implementation delays.

BENEFIT

Short term

- Immediate environmental safety by removing aging infrastructure.

Long term

- Cost savings in the long-run (no maintenance and safety cost of aging infrastructure).
- Existing pipelines could be cleaned and sold for reuse, providing potential cost offsets.

Short term

- Stakeholder engagement, Renewable energy source development that aligns with global decarbonisation efforts, Reduced costs compared to building new facilities.

Long term

- Aligns with Shell's sustainability goals and 2050 carbon neutrality targets, Position Shell in emerging green hydrogen market, Diversifies energy portfolio, reduced fossil fuel risks.

ENVIRONMENTAL IMPACT

- Risk of habitat loss during dismantling
- Potential disruption to ecosystems

- Reduced carbon emissions
- Preserved marine habitats
- Supports Biodiversity

STRATEGIC RECOMMENDATION FOR SHELL

- Repurposing as a Viable Strategy
- Alignment with Sustainability Goals
- Flexibility and Adaptation

industry examples: 1) BP's Hydrogen Pilot Project (UK), 2) Equinor's Hywind Project (Norway), 3) Perenco's decommission sites (UK)



STRATEGIC SOLUTION



Financial

Repurposing platforms for green hydrogen can deliver positive returns (NPV) over 15-20 years, aligning with sustainability goals



Environmental

Shell offsets decommissioning costs with renewable energy projects like offshore wind farms (AUD 146-223/MWh), generating revenue while lowering expenses

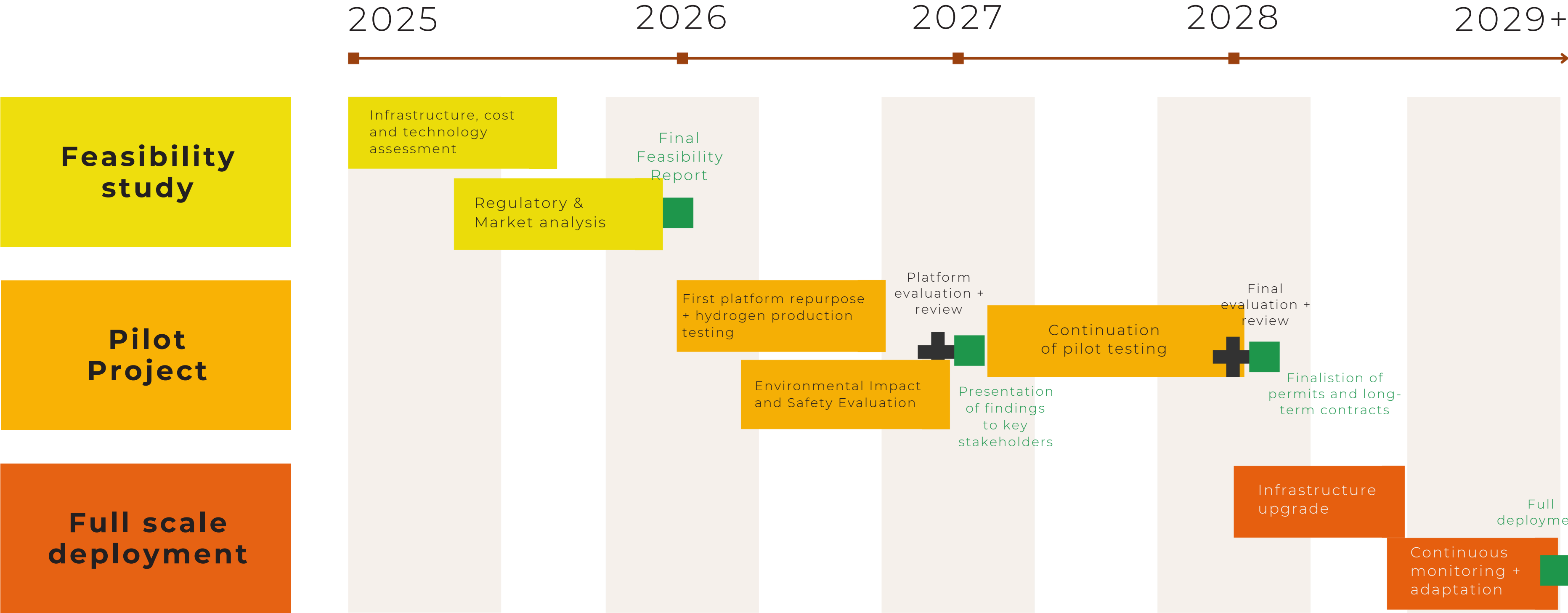


Regulatory

Shell will comply with UK and EU regulations, ensuring smooth approvals through Environmental Impact Assessments (EIA) and global environmental standards

IMPLEMENTATION PLAN

Shell plc UK could choose to implement partial decommissioning and repurposing to produce hydrogen using existing offshore windfarms.





IMPLEMENTATION PLAN: STAKEHOLDER ENGAGEMENT



Government & Regulators

Align Shell's projects with regulatory frameworks to secure approvals and support national energy transition policies



Local communities

Engage communities by promoting economic benefits like job creation and environmental initiatives such as marine conservation



Investors & Shareholders

Demonstrate long-term financial returns from sustainable energy projects that meet ESG goals and provide transparent performance tracking



RISK & OPPORTUNITIES

Shell plc UK may face certain risks and opportunities in the future market, which could be addressed with proper risk mitigation plans.



Key Factors

Risk & Opportunities

Risk Mitigation

- 1Regulatory Uncertainty
- 2Market demand for hydrogen
- 3Competition & market saturation

Risk: Evolving landscape for hydrogen generation industry with potential for changes in compliance requirements.
Opportunity: UK Hydrogen Strategy with potential for grants and subsidies.

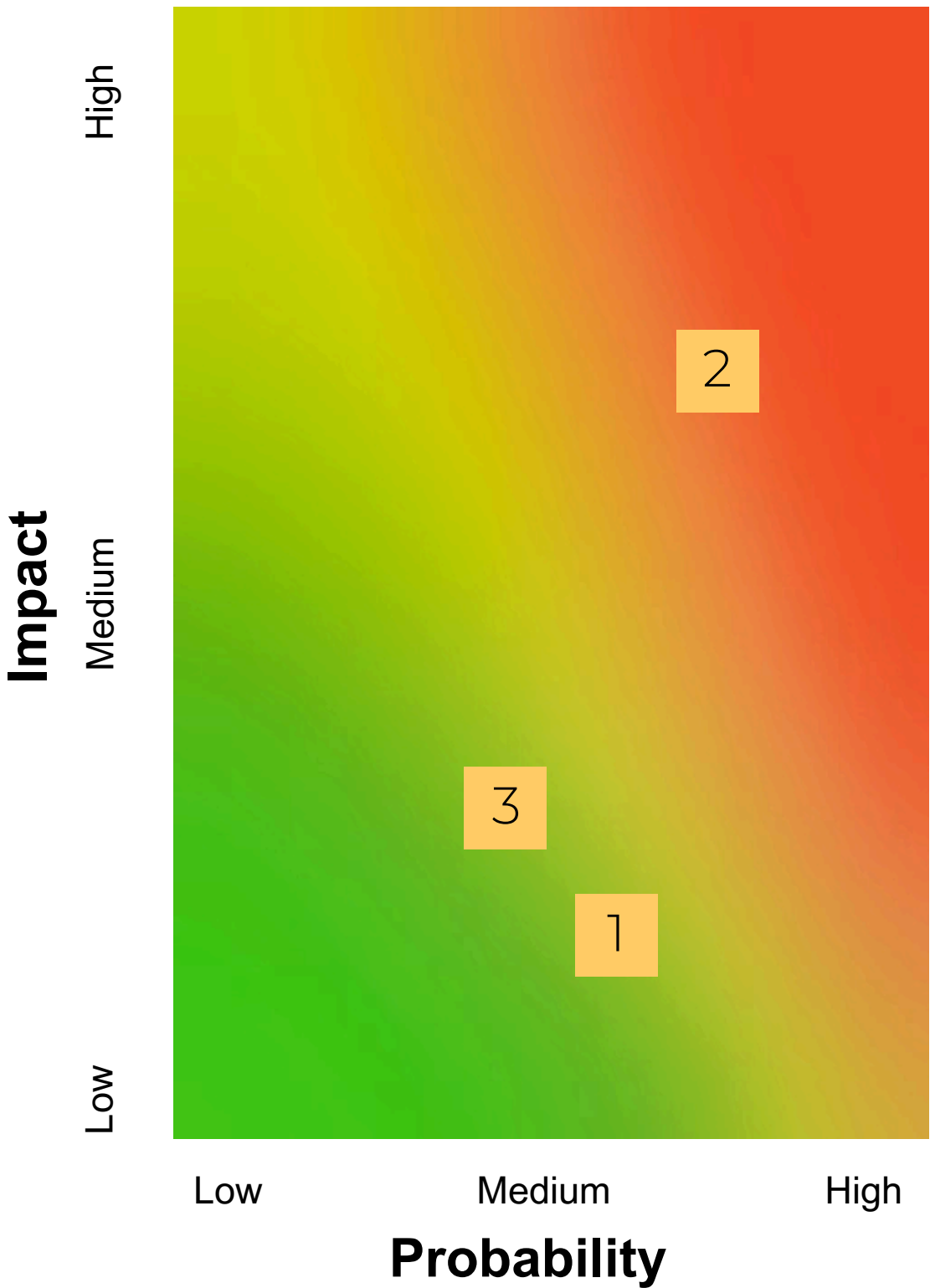
Risk: Uncertainty regarding how quickly end-use of hydrogen energy will be adopted.
Opportunity: Hydrogen plays a crucial role in decarbonizing industries where electrification isn't feasible.

Risk: Increased competition as other companies begin doing the same thing, potentially oversaturating the market.
Opportunity: First-mover advantage in the North Sea region, providing a competitive advantage. Allows Shell plc UK to establish itself as a leader.

Engage early with regulators to ensure project aligns with upcoming policies/standards and stay ahead.

Securing long-term agreements with industries or companies likely to adopt green hydrogen, such as shipping, aviation, chemicals and power generation.

Focus on continuous innovation and form strategic alliances to develop new technology and infrastructure, increasing market influence.



Sources: 1) [Hydrogen Strategy Update, August 2023](#), 2) [Demand for hydrogen](#)



CONCLUSION

Recommendation: Partial decommissioning and repurpose to produce green hydrogen (with integration of renewable electricity from offshore wind farms).

How can Shell help shape the markets of the future given its unique global position/scale?

Shell is faced with complex challenges in decommissioning its offshore platforms, including high costs, environmental risks, and strict regulatory requirements. However, by choosing to repurpose these assets for renewable energy solutions like green hydrogen and offshore wind, Shell can not only offset decommissioning costs but also align with global sustainability goals. Our strategy balances financial, environmental, and regulatory concerns, offering a path toward long-term profitability and leadership in the energy transition. By engaging key stakeholders—governments, local communities, and investors—Shell can secure the necessary support to shape the future of clean energy and drive impactful change in the industry.

APPENDIX

Cleaning of pipes for reuse

https://reports.shell.com/sustainability-report/2023/_assets/downloads/sog-sustainability-oil-gas-shell-sr23.pdf

Ecological considerations

<https://www.sciencedirect.com/science/article/pii/S0048969718350551?via%3Dihub>

Hydrogen Strategy Update, August 2023

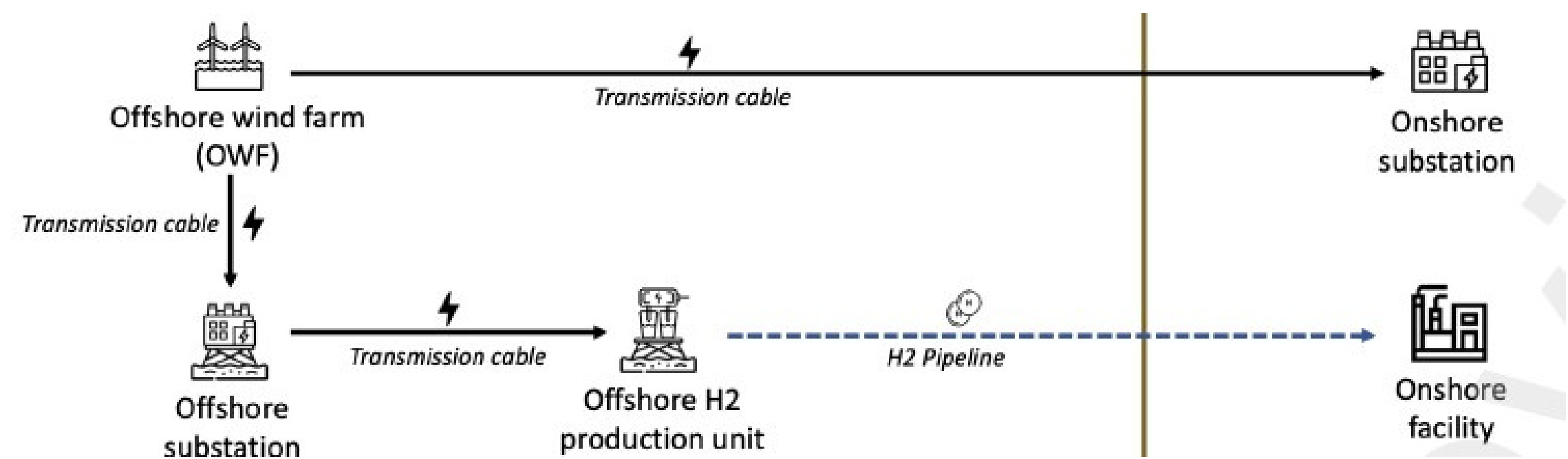
<https://assets.publishing.service.gov.uk/media/64e36b294002ee000d560c9f/hydrogen-strategy-update-to-the-market-august-2023.pdf>

Demand for hydrogen

<https://www.ft.com/content/327a6a16-2bb1-466e-8bd8-cd63b43b0a4b>



A study presented at SPE Europe Energy Conference and Exhibition, Italy, June 2024



**NPV: £132.6M, Payback
Period: 15.7 years.**

Scenario: Offshore green hydrogen production on a repurposed platform connected to a nearby wind farm.

Key Focus: Net Present Value (NPV) & payback periods over 20 years.

Assumed Hydrogen Price: £6.08/kg.

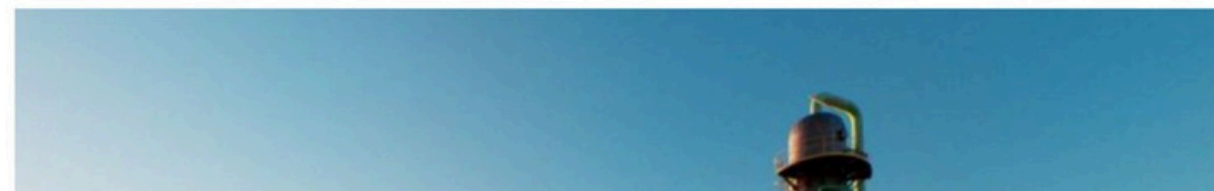
<https://onepetro.org/SPEEURO/proceedings-abstract/24EURO/1-24EURO/546514>

TECHNOLOGY | CCS/CCUS | HYDROGEN

Technip Awarded FEED for BP's UK Hydrogen Project

BP's H2Teesside is expected to be one of the U.K.'s largest low-carbon hydrogen production facilities and fully integrated with carbon capture technology.

Hart Energy Staff Tue, 08/27/2024 - 01:19 PM



BP's Hydrogen Pilot Project (UK)

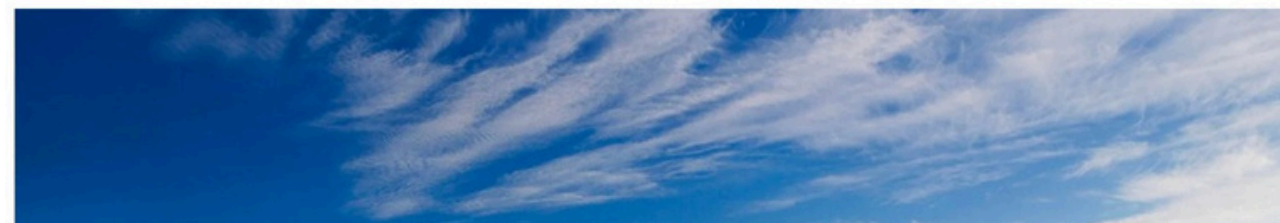
News |

Equinor launches world's largest floating wind farm in Norway

The 88MW project will power five Equinor oil and gas fields in the North Sea.

August 24, 2023

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Equinor's Hywind Project (Norway)

Perenco's decommission sites (UK)

Environment

🕒 This article is more than 2 months old

North Sea oil and gas firm Perenco failing to seal old wells, documents show

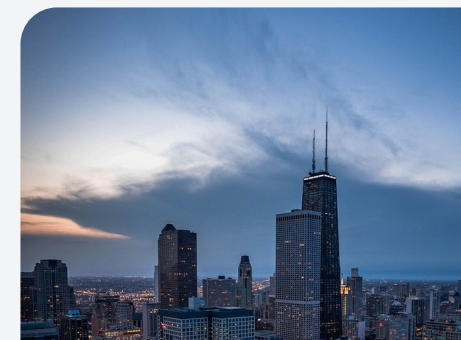
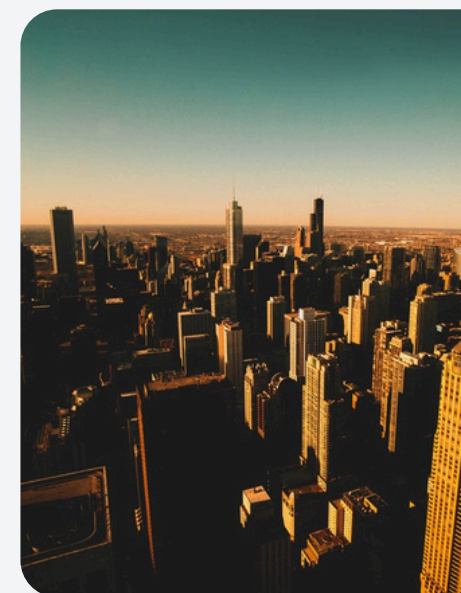
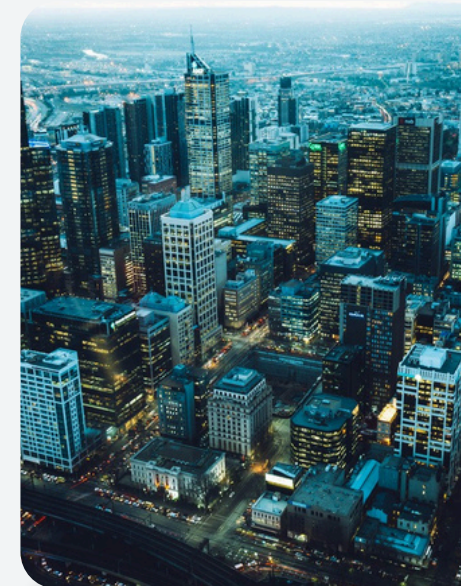
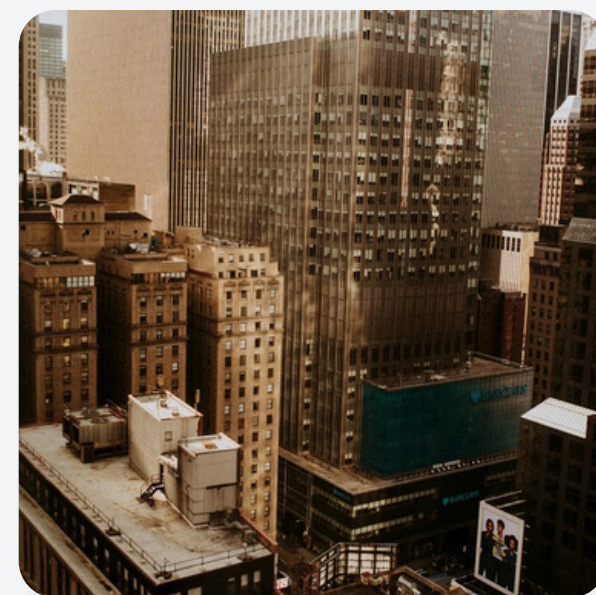
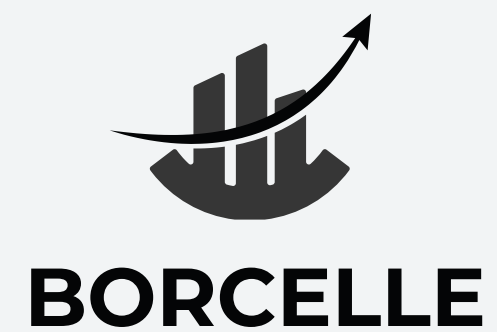
Fears of fire and environmental disaster as company repeatedly misses UK deadlines to decommission sites



Industry Examples and Influences (Slide 4)

APPENDIX: DETAILED IMPLEMENTATION PLAN

IMPLEMENTATION PLAN		2025				2026				2027				2028				2029+
		Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	
Feasibility Study																		
	Assess structural integrity of platforms and ability to house electrolysis technology																	
	Proximity and availability of offshore windfarms																	
	Condition of existing pipelines																	
	Estimate capital expenditure and operational cost of repurposing																	
	Research into companies with successful repurposing of oil rigs																	
	Evaluate cost-effectiveness of project																	
	Permit and license research																	
	Engage with relevant parties to ensure compliance with health, safety and environmental regulations																	
	Potential subsidies/grants																	
	Final feasibility report																	
Pilot Project																		
	Partner with nearby offshore wind farms to channel renewable energy																	
	First platform repurposing																	
	Begin small-scale production of green hydrogen																	
	Tracking of key performance indicators (energy efficiency, hydrogen yield, LCOH)																	
	Test hydrogen storage and transport methods using existing or upgraded pipelines																	
	Assess environmental impacts of production of marine ecosystem																	
	Conduct safety test, including hydrogen leak detections, integrity of pipelines																	
	Mid-term pilot testing review																	
	Presentation of findings to key stakeholders to secure funding																	
	Secure long-term partnerships with industries/companies with green hydrogen demand																	
	Continuation of pilot testing																	
	Final pilot test evaluation & review																	
	Finalisation of permits, long-term contracts and subsidies																	
Full Scale Deployment																		
	Begin repurposing of viable platforms (electrolysis unit installations, upgrading pipelines and OWF connections)																	
	Start decommissioning platforms deemed non-viable for hydrogen production																	
	Implement advanced hydrogen storage solutions and optimize hydrogen transport routes.																	
	Ensure full compliance with environmental and safety regulations																	
	Set up system for continuous monitoring																	
	Invest into R&D for new technologies or infrastructures																	
	Monitor hydrogen market to assess changes in demand																	



RESOURCE PAGE

